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7. (a) Crude oil can be separated into simpler mixtures, called fractions. Fractions contain hydrocarbon compounds called alkanes.

The table shows information about some of the fractions obtained from crude oil.

Fraction	Boiling point range (°C)	Number of carbon atoms in the alkanes
refinery gases	−160 to 20	1–4
petrol	20 to 240	4–12
naphtha	100 to 250	7–14
kerosene	200 to 280	11–15
diesel oil	280 to 350	15–19

Use only the information in the table to answer parts (i)–(iii).

(i) Complete the sentence below by <u>underlining</u> the correct word(s). [1]

As the number of carbon atoms in an alkane increases,

the boiling point (**increases** / **decreases** / **stays the same**).

(ii) Hexane has a boiling point of 69 °C.

Give the name of the fraction that contains hexane. [1]

....

(iii) One alkane is found in the kerosene and diesel oil fractions.

Give the number of carbon atoms in this alkane. [1]

....

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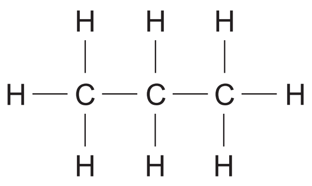
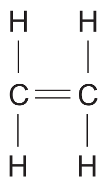
(b) Diagrams **A–E** show the structural formulae of some carbon compounds.



A



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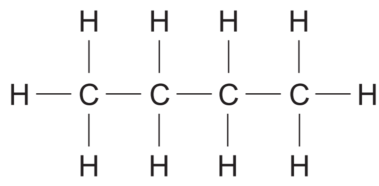

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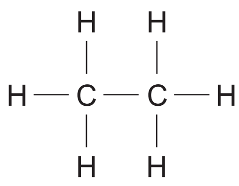
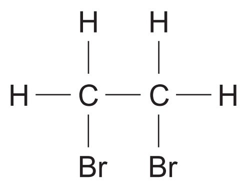


D



E

(i) Circle the molecular formula for compound **A**. [1]

$\mathbf{C_3H_3}$ $\mathbf{C_8H_3}$ $\mathbf{C_3H_8}$ **3C8H** $\mathbf{C^3H^8}$

(ii) Give the **letter** of the compound that is unsaturated. [1]

....

(iii) Explain why compound **E** is **not** a hydrocarbon. [1]

....

....

(iv) Give the **letter** of the compound that belongs to the family of carbon compounds with the general formula C_nH_{2n}. [1]

....

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(c) The chart shows the stages used in the laboratory to make ethanol from glucose solution.

Stage **1**		Stage **2**		Stage **3**
yeast is added to glucose solution and left in a warm place until the reaction stops	→	yeast is removed from the reaction mixture	→	ethanol is separated from the reaction mixture

(i) Give the number of the stage in which distillation is used. [1]

Stage ....

(ii) The formula of ethanol is C_2H_5OH.

Complete the structure of ethanol by adding all the atoms and bonds present. [1]

C — C

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(iii) Ethanol is the alcohol found in alcoholic drinks.

The following information appears on a bottle of wine.

Volume	750 millilitres
Alcohol by volume	13.5%

The formula below can be used to calculate the mass of ethanol in an alcoholic drink.

mass of ethanol (g) = 8 × volume (litres) × alcohol by volume (%)

Use the formula to find the mass of ethanol in a bottle of wine. [2]

1 litre = 1000 millilitres

Mass = ... g

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